

$^{36}\text{Ar}(\text{p}, ^3\text{He})$ **1969Br21**

$^{36}\text{Ar} \rightarrow \text{p} + \text{n} + ^{34}\text{Cl}$ from $J^\pi=0^+$ ^{36}Ar ground state.

1969Br21: a 45-MeV proton beam was produced from the Berkeley 88-Inch Cyclotron. The target was 99.6%-enriched ^{36}Ar gas.

Reaction products were detected using a 155- μm phosphorus-diffused silicon ΔE transmission counter and a 3.0-mm lithium-drifted silicon E counter operated in coincidence, followed by a 0.5-mm lithium-drifted silicon E-reject counter operated in anti-coincidence with the first two counters to eliminate long-range particles. Measured $\sigma(E(^3\text{He}), \theta)$. Deduced levels, L-transfers, J, π , and isospin. Also measured $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ and $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$.

 ^{34}Cl Levels

Selection rules for direct two-particle transfer reactions: $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ transfers particles with $T=1$, $S=0$; $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ transfers particles with $T=0$, $S=1$. $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ transfers particles with $T=1$, $S=0$ if $\sigma(\theta)$ of $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ matches $\sigma(\theta)$ of the mirror state in ^{34}Ar populated by $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ and absent in $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$.

E(level)	J^π	$L^\ddagger$	Comments
0.0	0^+	0	$T=1$
460 [†]			$T=0$
1231 [†]			$T=0$
			weakly populated (1969Br21).
1891 [†]			$T=0$
			weakly populated (1969Br21).
2162 [†]	2^+	2	$T=1$
2600 30			$T=0$
3130 30			$T=0$
3350 50	2^+	2	$T=1$
3940 40	0^+	0	$T=1$
4670 40	(3^-)	(3)	$T=1$
4970 40		0	E(level): Doublet of $T=0$ and 1. $T=1$ component corresponds to 0^+ .
5600 40			$T=0$
6160 40		2	E(level): Doublet of $T=0$ and 1. $T=1$ component corresponds to 2^+ .
7070 40			$T=0$

[†] Observed in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ reaction and quoted E(level) from **1967En05** without reporting their own values (**1969Br21**).

[‡] From DWBA analysis of mirror $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ (**1969Br21**).